

Ajay V V

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EDUCATION

Hindusthan College Of Engineering and Technology

Bachelor of Technology in Artificial Intelligence and Machine Learning

Coimbatore, Tamil Nadu

Expected May 2028

TECHNICAL SKILLS

Languages: Python, C++, SQL

AI/ML & Vision: TensorFlow, Keras, OpenCV, Scikit-learn, YOLOv8, Deep Learning

Developer Tools: Git, GitHub, Google Colab, VS Code

Core Concepts: Data Structures & Algorithms (DSA), Data Preprocessing, Model Evaluation

PROJECTS

Industrial Safety Monitor | *Python, OpenCV, YOLOv8, Google Colab*

- Developed a real-time computer vision system to detect safety equipment compliance in workspace environments.
- Curated and annotated a custom dataset, applying bounding-box data augmentations to improve detection accuracy.
- Trained a lightweight YOLOv8 model to achieve fast inference speeds suitable for live video streams.

Voice-Shield (Audio Deepfake Classifier) | *Python, TensorFlow, Keras, Librosa*

- Engineered a deep learning classifier to distinguish authentic human speech from AI-generated synthetic audio.
- Extracted audio features (MFCCs, spectrograms) from raw sound files for input into deep neural networks.
- Evaluated model performance using precision-recall metrics to minimize false positives in synthetic voice detection.

QR-Shield (Quishing Defender) | *Python, Image Processing, Security Logic*

- Built a specialized desktop security utility to parse and decode static QR code image files.
- Implemented URL extraction routines to inspect encoded payload destinations prior to client browser redirects.
- Structured the codebase for modular execution and published documentation on GitHub.